TTLD-Net: Topology-Aware Tiny Traffic Light Detection Network via Implicit Topology Learning

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Abstract

Tiny Traffic Light Detection remains a challenging problem in autonomous driving due to the extremely small size of traffic lights in high-resolution images. Existing detectors primarily rely on local appearance features and often struggle with feature degradation, false positives, and poor domain generalization. We propose **Topology-Aware Tiny Traffic Light Detection Network (TTLD-Net)**, a novel two-stage framework that combines high-recall candidate generation, implicit topology reasoning, and verification learning. Unlike previous approaches that depend solely on object appearance, TTLD-Net learns the latent spatial topology surrounding a traffic light candidate through a deformable attention-based context reasoning mechanism. The proposed method automatically discovers structural cues without requiring explicit semantic annotations such as poles, roads, or intersections. Experimental results are expected to demonstrate significant improvements in AP_{small}, Recall, and False Positive reduction compared with state-of-the-art detectors.

1 Introduction

Traffic light detection is a fundamental perception task in autonomous driving and intelligent transportation systems. In real-world datasets such as the Bosch Small Traffic Lights Dataset, traffic lights often occupy only 8×8 pixels within images of several megapixels.

This introduces three major challenges:

- **Feature Degradation:** Tiny objects lose discriminative information after multiple down-sampling operations in deep neural networks.
- **Appearance Ambiguity:** Small traffic lights visually resemble vehicle brake lights, LED billboards, and reflective signs, leading to severe false positives.
- **Domain Shift:** Generalization fails across day/night or camera systems when relying purely on appearance.

To address these issues, TTLD-Net introduces topology-aware reasoning based on latent spatial relationships rather than handcrafted geometric rules or explicit object annotations.

2 Overall Architecture

The framework first generates high-recall candidates by preserving ambiguous detections, then reasons over their surrounding spatial context using dynamic feature sampling, and finally verifies the topology validity. The architecture is illustrated in Fig. 1.

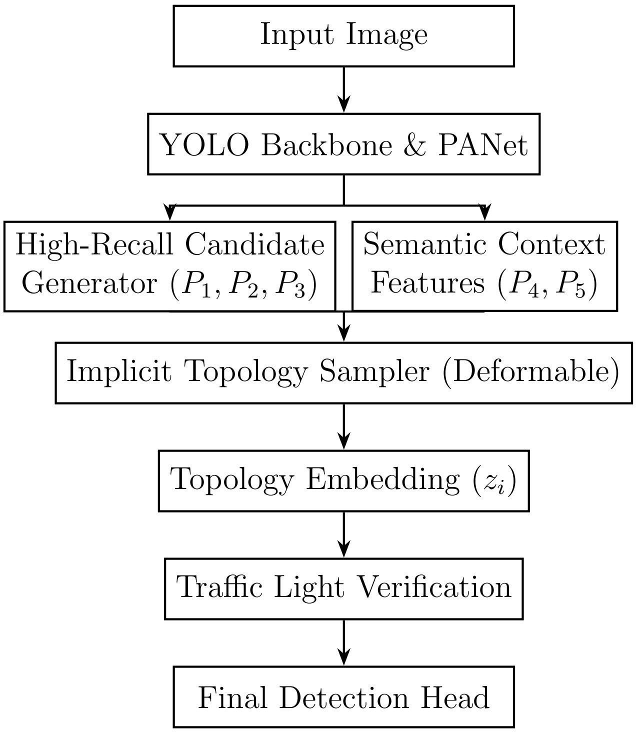

Figure 1: Overall architecture of TTLD-Net. The network utilizes parallel pathways for candidate localization and context extraction before aggregating them through an implicit topology sampler.

3 High-Recall Tiny Candidate Generator

The objective of this module is to produce traffic light candidates $\mathcal{C} = \{c_1, c_2, \ldots, c_N\}$ with extremely high recall from high-resolution feature maps $\mathcal{F}_{shallow} = \{P_1, P_2, P_3\}$.

To preserve tiny, ambiguous traffic lights that are typically discarded by standard detectors, we fundamentally adjust the localization filtering mechanism:

1. **Extreme Low-Confidence Acceptance:** The objectness confidence threshold is significantly lowered ($\tau \approx 0.05$) to retain hard examples.

2. **Soft-NMS Integration:** Traditional Non-Maximum Suppression is replaced with Soft-NMS to prevent the erroneous suppression of spatially clustered small traffic lights.

3. **Focal Loss Tuning:** The focusing parameter γ and balancing factor α in the Focal Loss are dynamically tuned to penalize false negatives heavier than false positives during the primary localization phase.

This intentional generation of a noisy, high-recall candidate pool explicitly delegates the complex task of false positive suppression to the subsequent context-aware modules.

4 Implicit Topology Sampler via Deformable Attention

Instead of utilizing static cropped grids or explicit graph structures (which suffer from severe spatial misalignment across scales), TTLD-Net extracts latent context using a deformable attention mechanism.

4.1 Feature Subspace Projection

Candidate features f_{cand} and deep semantic context features $\mathcal{F}_{ctx} = \{P_4, P_5\}$ are first projected into a unified d-dimensional subspace using 1×1 convolutions to resolve channel dimension inconsistencies prior to attention computation.

4.2 Dynamic Context Aggregation

For each candidate c_i, its center coordinates serve as the reference point p_q. The module dynamically predicts K sampling offsets Δp_{qk} and corresponding attention weights A_{qk} over the semantic feature map f_{ctx}.

The final **Topology Embedding** z_i is computed via:

$$z_i = \sum_{k=1}^{K} A_{qk} \cdot \mathbf{W} f_{ctx}(p_q + \Delta p_{qk}) \tag{1}$$

where $\mathbf{W}$ is a learnable projection weight matrix. The feature $f_{ctx}(p_q + \Delta p_{qk})$ is sampled at fractional spatial locations utilizing bilinear interpolation. This differential mechanism allows the network to implicitly "reach out" to contextual cues (e.g., poles, sky, crosswalks) without requiring explicit pixel-level annotations.

5 Traffic Light Verification Module

To eliminate the massive amount of false positives (e.g., vehicle brake lights, LED advertisements) generated in the first stage, a verification network fuses local and contextual cues:

$$v_i = [f_{cand} \parallel z_i] \tag{2}$$

where $\parallel$ denotes the concatenation operation. A lightweight Multi-Layer Perceptron (MLP) processes v_i to predict a verification score $P(\text{Valid})$. Candidates with verification scores below a strict threshold are subsequently discarded.

6 Loss Function and Contrastive Learning

The overall training objective is defined as:

$$L_{total} = L_{det} + \lambda_1 L_{topology} + \lambda_2 L_{verify} \tag{3}$$

where L_{det} represents the standard YOLO detection loss (bounding box regression and state classification), and L_{verify} is the Binary Cross-Entropy loss for the verification head.

6.1 Topology Contrastive Loss

To ensure the Topology Embedding z_i strictly encodes meaningful geometric structures rather than random noise, we introduce an **InfoNCE-based Contrastive Loss**.

Let z_i be the topology embedding of a valid candidate. We define z_{ctx}^{+} as the embedding of the true surrounding context (positive pair) and $\{z_{ctx,j}^{-}\}_{j=1}^{M}$ as a set of randomly sampled contexts from other regions or images in the mini-batch (negative pairs). The topology loss is formulated as:

$$L_{topology} = -\log \frac{\exp(\text{sim}(z_i, z_{ctx}^{+})/\tau)}{\exp(\text{sim}(z_i, z_{ctx}^{+})/\tau) + \sum_{j=1}^{M} \exp(\text{sim}(z_i, z_{ctx,j}^{-})/\tau)} \tag{4}$$

where $\text{sim}(u, v) = \frac{u^T v}{\|u\|\|v\|}$ denotes the cosine similarity, and τ is the temperature hyperparameter. This objective explicitly forces the model to construct a latent space where false positive contexts are pushed far away from the valid traffic-light manifold.

7 Scientific Contributions

- Propose TTLD-Net, a topology-aware hybrid framework for extreme tiny traffic light detection.
- Introduce an Implicit Topology Sampler utilizing Deformable Attention, mitigating feature degradation and eliminating the need for hard-coded rules or explicit scene annotations.
- Design a Topology Contrastive Objective (InfoNCE) to effectively separate true traffic lights from visually similar distractors in the latent space.

8 Expected Improvements

Compared with state-of-the-art detectors such as YOLO variants and RT-DETR, TTLD-Net is expected to yield the performance improvements detailed in Table 1. The framework specifically addresses the precision-recall trade-off by drastically reducing the false positive rate while maintaining high recall for sub-10 pixel objects.

Table 1: Expected Performance Gains compared with appearance-based baseline models.

Metric	**Expected Gain**
AP_{50}	$+1.5\% \sim +3.0\%$
AP_{small}	$+3.0\% \sim +6.0\%$
Recall	$+2.0\% \sim +5.0\%$
Precision	$+5.0\% \sim +8.0\%$
False Positive Rate	$-15.0\% \sim -30.0\%$